PROJECT ON

# QUIZ GAME

**NAME:** RASHI KASHYAP
**COURSE:** BCA DATA SCIENCE
**SESSION:** 2025-26

# INTRODUCTION

This project focuses on developing a simple and interactive quiz application using the C++ programming language. The application is designed to run on a console and provides users with a series of multiple-choice questions. Each question includes four options, out of which only one is correct. The user selects an option, and the program checks whether the answer is right or wrong.

The main purpose of this project is to demonstrate how basic programming concepts can be applied to create a functional application. It uses object-oriented programming principles, where a class is defined to store questions, options, and correct answers. Functions are used to display questions, take input from the user, and calculate the score.

As the user progresses through the quiz, the program provides immediate feedback for each answer. At the end of the quiz, the total score is displayed, giving a clear idea of the user's performance. This project helps in understanding important C++

# OBJECTIVE

- To design and develop a simple quiz application using C++
- To understand and apply object-oriented programming concepts like classes and objects
- To practice taking user input and displaying output
- To implement decision-making using conditional statements
- To create an interactive program that evaluates user performance
- To improve basic programming and logical thinking skills

# FEATURES

- Simple and user-friendly console interface
- Multiple-choice questions with four options
- Immediate feedback (Correct/Wrong) after each answer
- Automatic score calculation
- Uses class to manage question data
- Easy to modify or add new questions

# SOURCE CODE: PART 1

```
#include<iostream>
using namespace std;
class Question {
public:
string ques, op1, op2, op3, op4; int correct;

void setQuestion(string q, string a, string b, string c, string d, int ans) {
ques = q; op1 = a; op2 = b; op3 = c; op4 = d; correct = ans;    }

        int askQuestion()
            { int choice;
cout << "\n" << ques << endl;
cout << "1. " << op1 << "\n2. " << op2 << "\n3. " << op3 << "\n4. " << op4 << endl;
cin >> choice;
if(choice == correct) { cout << "Correct\n"; return 1; }
else { cout << "Wrong\n"; return 0; }
        }
    };
```

# SOURCE CODE: PART 2

```
intmain() {
int score = 0; Question q1, q2, q3, q4, q5, q6, q7, q8;

q1.setQuestion("Who developed C++?", "Dennis", "Bjarne", "James", "George", 2);
q2.setQuestion("Output operator?", ">>", "<<", "==", "=", 2);    q3.setQuestion("File extension?", ".c", ".cpp", ".txt", ".java", 2);    q4.setQuestion("Keyword for class?", "class", "struct", "define", "obj", 1);    q5.setQuestion("Input function?", "cout", "cin", "print", "scan", 2);    q6.setQuestion("Loop with fixed count?", "for", "while", "do", "switch", 1);    q7.setQuestion("Decimal data type?", "int", "char", "float", "bool", 3);
q8.setQuestion("One name many forms?", "inherit", "encap", "poly", "abs", 3);

score += q1.askQuestion(); score += q2.askQuestion(); score += q3.askQuestion();
score += q4.askQuestion(); score += q5.askQuestion(); score += q6.askQuestion();
score += q7.askQuestion(); score += q8.askQuestion();

cout << "\nFinal Score = " << score << "/8" << endl; return0;
}
```

# Advantages and disadvantages

**Advantages**

Easy to understand and implement

Uses basic C++ and OOP concepts (good for learning)

Interactive (user gets instant feedback)

Automatic score calculation

Simple to modify or add questions

**Disadvantages**

Only console-based

Fixed questions

No timer or difficulty levels

No user data saving (score not stored)

Limited scalability (not suitable for large quizzes)

# CONCLUSION

This project demonstrates how a simple quiz application can be built using C++ by applying basic programming and object-oriented concepts. It shows how classes and functions help in organizing code and making it reusable. The program successfully takes user input, checks answers, and calculates the final score in an efficient way. Overall, this project improves understanding of C++ fundamentals and provides a base for developing more advanced and interactive applications in the future.

# THANK YOU !